



1. 1.2 Pre-eclampsia and HTN Disease of Pregnancy

Clamps on red pipes

→ preeclampsia = HTN ($\geq 140/90$) + proteinuria ($\geq 300\text{mg}/24\text{h}$ or $\text{UPC} \geq 0.3$)
 atier 20wks GA (MC time)
 → 3rd tri)

2. High pressure steam escaping from clamped red pipes

→ preeclampsia = HTN ($\geq 140/90$) + proteinuria ($\geq 300\text{mg}/24\text{h}$ or $\text{UPC} \geq 0.3$)
 atier 20wks GA (MC time)
 → 3rd tri)

3. Chicken drumsticks in spilled yellow drink next to clamped red pipes

→ preeclampsia = HTN ($\geq 140/90$) + proteinuria ($\geq 300\text{mg}/24\text{h}$ or $\text{UPC} \geq 0.3$)
 atier 20wks GA (MC time)
 → 3rd tri)

4. Reactor pressure gauges set to 140/90

→ HTN in pregnancy = $\text{SBP} \geq 140$ and/or $\text{DBP} \geq 90$ 2x at least 4h apart

5. Grandfather clock at 19:00 by pressure gauges

→ cHTN dx in pregnancy = HTN dx at $< 20\text{wks GA}$ OR HTN dx'd in pregnancy
 that lasts $> 12\text{wks}$
 postpartum, NO

6. Week 20 calendar next to pressure gauges

gestational HTN = HTN w/o proteinuria dx >20wks GA, NO proteinuria, resolves by 12wks

postpartum

7. “-3” anchor credit card pays for chicken

→ proteinuria in pregnancy = >300mg/24h on 24h urine protein collection OR
urine
protein:creatinine (UPC) ≥ 0.3

8. 3-shaped anchor on submarine

→ preeclampsia = HTN ($\geq 140/90$) + proteinuria (>300mg/24h or UPC ≥ 0.3)
after 20wks GA (MC time
→ 3rd tri)

9. Spiral red pipes

→ normal placental development = cytotrophoblasts invade maternal spiral
arteries to transform
them from high resistance → low resistance vessels

10. Patches on spiral red pipes

→ preeclampsia pathology = abnormal remodeling of the spiral arteries
(placental vessels remain
high resistance) → placental hypoperfusion/ischemia → protein release →
endothelial dysfunction
→ vasospasm (HTN) + vascular permeability (proteinuria)

11. Spiral red pipes releasing hazardous particles

→ preeclampsia pathology = abnormal remodeling of the spiral arteries
(placental vessels remain
high resistance) → placental hypoperfusion/ischemia → protein release →
endothelial dysfunction
→ vasospasm (HTN) + vascular permeability (proteinuria)

12. Rusty red pipe

→ preeclampsia pathology = abnormal remodeling of the spiral arteries
(placental vessels remain
high resistance) → placental hypoperfusion/ischemia → protein release →
endothelial dysfunction
→ vasospasm (HTN) + vascular permeability (proteinuria)

13. Fluid leaking from red pipe

→ preeclampsia pathology = abnormal remodeling of the spiral arteries
(placental vessels remain
high resistance) → placental hypoperfusion/ischemia → protein release →
endothelial dysfunction
→ vasospasm (HTN) + vascular permeability (proteinuria)

14. Narrowed portions of red pipe

→ preeclampsia pathology = abnormal remodeling of the spiral arteries
(placental vessels remain
high resistance) → placental hypoperfusion/ischemia → protein release →
endothelial dysfunction
→ vasospasm (HTN) + vascular permeability (proteinuria)

“ALERT! SEVERE REACTOR DAMAGE!” sign

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preeclampsia with severe features (has signs of end organ damage) = proteinuria + 1 of the

following → 1. Sustained severe range BPs (SBP $\geq$ 160 and/or DBP $\geq$ 110 2x at least 4h apart) 2. Lab

criteria (1 or more) = plts <100,000, AST/ALT 2x ULN, Cr >1.1 3. Clinical criteria (1 or more) =

pulmonary edema, persistent RUQ/epigastric pain, HA not responsive to meds, changes in vision

Show more

16. Sailor shaking with fear

→ eclampsia = preeclampsia + seizures

17. Captain w/ wet lung lapels

→ maternal complications of preeclampsia = 1. Pulmonary edema 2. Hemorrhagic stroke 3. DIC 4. AKI → renal failure 5. Liver injury

18. Swipe of blood on captain's hat

→ maternal complications of preeclampsia = 1. Pulmonary edema 2. Hemorrhagic stroke 3. DIC 4. AKI → renal failure 5. Liver injury

19. DICe badge on captain

→ maternal complications of preeclampsia = 1. Pulmonary edema 2. Hemorrhagic stroke 3. DIC 4. AKI → renal failure 5. Liver injury

20. Broken kidney microphone held by captain

→ maternal complications of preeclampsia = 1. Pulmonary edema 2. Hemorrhagic stroke 3. DIC 4. AKI → renal failure 5. Liver injury

21. Liver shaped rupture in captain's uniform

→ maternal complications of preeclampsia = 1. Pulmonary edema 2. Hemorrhagic stroke 3. DIC 4. AKI → renal failure 5. Liver injury

22. Small crouching sailor underneath desk

→ fetal complications of preeclampsia = 1. IUGR 2. Oligohydramnios 3. IUFD 4. Placental abruption 5. Non-reassuring fetal heart tracing 6. Preterm birth

23. Little puddle underneath small sailor

→ fetal complications of preeclampsia = 1. IUGR 2. Oligohydramnios 3. IUFD 4. Placental abruption 5. Non-reassuring fetal heart tracing 6. Preterm birth

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Pirate ship with skull flag

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fetal complications of preeclampsia = 1. IUGR 2. Oligohydramnios 3. IUFD 4. Placental abruption 5.

Non-reassuring fetal heart tracing 6. Preterm birth

25. Placenta shaped speaker rupturing out of the wall

→ fetal complications of preeclampsia = 1. IUGR 2. Oligohydramnios 3. IUFD 4. Placental abruption 5.

Non-reassuring fetal heart tracing 6. Preterm birth

26. Route tracing map

→ fetal complications of preeclampsia = 1. IUGR 2. Oligohydramnios 3. IUFD 4. Placental abruption 5.

Non-reassuring fetal heart tracing 6. Preterm birth

27. Premature missile launch!

→ fetal complications of preeclampsia = 1. IUGR 2. Oligohydramnios 3. IUFD 4. Placental abruption 5.

Non-reassuring fetal heart tracing 6. Preterm birth

28. Family portrait of a grandma and a baby

→ preeclampsia demographics = extremes of maternal age, BMI>25, African American, nulliparous, multifetal gestations

29. "PVT: P0" badge

→ preeclampsia demographics = extremes of maternal age, BMI>25, African American, nulliparous, multifetal gestations

30. Multiple buns

→ preeclampsia demographics = extremes of maternal age, BMI>25, African American, nulliparous, multifetal gestations

31. Enlarged waistline

→ preeclampsia demographics = extremes of maternal age, BMI>25, African American, nulliparous, multifetal gestations

32. African-American sailor

→ preeclampsia demographics = extremes of maternal age, BMI>25, African American, nulliparous, multifetal gestations

33. New high pressure leak from a red pipe

→ preeclampsia pt. presentation = new onset HTN ($\geq 140/90$), HA/changes in vision/RUQ or epigastric pain

34. Sailor who hit his head
→ preeclampsia pt. presentation = new onset HTN ($\geq 140/90$), HA/changes in vision/RUQ or epigastric pain
35. Flashing light + smoke impairing sailor's vision
→ preeclampsia pt. presentation = new onset HTN ($\geq 140/90$), HA/changes in vision/RUQ or epigastric pain
36. Debris hitting sailor's stomach
→ preeclampsia pt. presentation = new onset HTN ($\geq 140/90$), HA/changes in vision/RUQ or epigastric pain
37. EMERGENCY sign next to high pressure pipes
→ HTNsive emergency = persistent severe HTN (SBP ≥ 160 and/or DBP ≥ 110)
38. History of Naval Warfare book
→ RF for preeclampsia = 1. h/o preeclampsia 2. cHTN 3. DM 4. Renal dz. 5. SLE/APLS 6. FHx of preX
39. Clock next to high pressure leak
→ RF for preeclampsia = 1. h/o preeclampsia 2. cHTN 3. DM 4. Chronic renal dz 5. SLE/APLS 6. FHx of preX
40. Bowl of candy
→ RF for preeclampsia = 1. h/o preeclampsia 2. cHTN 3. DM 4. Chronic renal dz 5. SLE/APLS 6. FHx of preX
41. Wrinkled kidney socks next to grandfather clock
→ RF for preeclampsia = 1. h/o preeclampsia 2. cHTN 3. DM 4. Chronic renal dz. 5. SLE/APLS 6. FHx of preX
42. USS Seawolf submarine + antibody antenna snagging a phospholipid jellyfish
→ RF for preeclampsia = 1. h/o preeclampsia 2. cHTN 3. DM 4. Chronic renal dz. 5. SLE/APLS 6. FHx of preX
43. Family photo next to sailor w/ preeclampsia sx
→ RF for preeclampsia = 1. h/o preeclampsia 2. cHTN 3. DM 4. Chronic renal dz. 5. SLE/APLS 6. FHx of preX

44. Wet lung life-vest with crackly sounding velcro
→ preeclampsia PE = 1. Lung exam → check for crackles (pulmonary edema) 2. Neuro exam → check for neuro deficits (c/f stroke) 3. MSK exam → edema (2/2 low oncotic pressure)
45. Brain patterned naval hat
→ preeclampsia PE = 1. Lung exam → check for crackles (pulmonary edema) 2. Neuro exam → check for neuro deficits (c/f stroke) 3. MSK exam → edema (2/2 low oncotic pressure)
46. Baggy pants dripping water
→ preeclampsia PE = 1. Lung exam → check for crackles (pulmonary edema) 2. Neuro exam → check for neuro deficits (c/f stroke) 3. MSK exam → edema (2/2 low oncotic pressure)
47. Mop handle in yellow bucket
→ ways to assess for proteinuria in pregnancy = 1. Urine dipstick → $\geq 2+$ protein 2. Urine protein:creatinine ratio (UPC) ≥ 0.3 3. 24h urine protein collection → $\geq 300\text{mg}/24\text{h}$
48. “-·3” anchor credit card pays for chicken
→ ways to assess for proteinuria in pregnancy = 1. Urine dipstick → $\geq 2+$ protein 2. Urine protein:creatinine ratio (UPC) ≥ 0.3 3. 24h urine protein collection → $\geq 300\text{mg}/24\text{h}$
49. Falling plates
→ preeclampsia lab w/u = 1. CBC → check if $\text{plts} < 100,000/\text{mL}$ 2. Cr → check if $\geq 1.1\text{mg}/\text{dL}$ 3. AST/ALT → check if 2x ULN
50. “ $\geq 1.1\%$ APR” Credit card
→ preeclampsia lab w/u = 1. CBC → check if $\text{plts} < 100,000/\text{mL}$ 2. Cr → check if $\geq 1.1\text{mg}/\text{dL}$ 3. AST/ALT → check if 2x ULN
51. 2 ToASTed mALT beers up high on a table
→ preeclampsia lab w/u = 1. CBC → check if $\text{plts} < 100,000/\text{mL}$ 2. Cr → check if $\geq 1.1\text{mg}/\text{dL}$ 3. AST/ALT → check if 2x ULN
52. DDx menu
→ DDx for HTN in pregnancy = chronic HTN, gestational HTN, preeclampsia, preeclampsia w/ severe features, superimposed preeclampsia on chronic HTN, HELLP, eclampsia
53. Sailor underneath HELLP! Sign vomiting

HELLP Syndrome pt. presentation = nausea/vomiting, RUQ/epigastric pain 2/2 liver capsule

stretching (complication → liver rupture → hemorrhage)

54. Sailor underneath HELLP! Sign with ruptured, bleeding shirt in the shape of a liver

→ HELLP Syndrome pt. presentation = nausea/vomiting, RUQ/epigastric pain 2/2 liver capsule

stretching (complication → liver rupture → hemorrhage)

55. Pirate evilly laughing “MAHA-hahaha”

→ HELLP Syndrome lab findings = 1. Microangiopathic hemolytic anemia (↑ bilirubin, ↑ LDH, ↓ haptoglobin, smear → schistocytes) 2. ↑ ↑ AST/ALT (2x ULN) 3. Thrombocytopenia (<100,000)

56. Pirate wearing a red helmet

→ HELLP Syndrome lab findings = 1. Microangiopathic hemolytic anemia (↑ bilirubin, ↑ LDH, ↓ haptoglobin, smear → schistocytes) 2. ↑ ↑ AST/ALT (2x ULN) 3. Thrombocytopenia (<100,000)

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→ HELLP Syndrome lab findings = 1. Microangiopathic hemolytic anemia (↑ bilirubin, ↑ LDH, ↓ haptoglobin, smear → schistocytes) 2. ↑ ↑ AST/ALT (2x ULN) 3. Thrombocytopenia (<100,000)

58. Falling plates

→ HELLP Syndrome lab findings = 1. Microangiopathic hemolytic anemia (↑ bilirubin, ↑ LDH, ↓ haptoglobin, smear → schistocytes) 2. ↑ ↑ AST/ALT (2x ULN) 3. Thrombocytopenia (<100,000)

59. Baby submarine with FHR tracing pattern on the side

→ maternal preeclampsia's effect on fetal physiology → placental hypoperfusion. To monitor for fetal well-being → antenatal testing (biophysical profile, non-stress test), serial growth ultrasounds

60. Baby submarine emitting sonar waves

→ maternal preeclampsia's effect on fetal physiology → placental hypoperfusion. To monitor for fetal well-being → antenatal testing (biophysical profile, non-stress test), serial growth ultrasounds

61. Navy Seal inside the water with a severe look on his face holding a HELLP sign

→ hypertensive diseases of pregnancy that should be managed as an inpatient = preeclampsia w/ SF, HELLP, eclampsia. Monitoring → inpatient BP checks, antenatal testing until delivery

62. Alpha and Beta divers on a boat

→ BP medications to use in hypertensive diseases of pregnancy = 1. IV/PO labetalol (don't use if HR <60bpm) 2. IV/PO hydralazine 3. PO nifedipine

Hydro boat

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BP medications to use in hypertensive diseases of pregnancy = 1. IV/PO labetalol (don't use if HR

<60bpm) 2. IV/PO hydralazine 3. PO nifedipine

64. Diver with a KNIFE

→ BP medications to use in hypertensive diseases of pregnancy = 1. IV/PO labetalol (don't use if HR

<60bpm) 2. IV/PO hydralazine 3. PO nifedipine

65. 150/100 compass

→ goal of BP management in hypertensive diseases of pregnancy = 130-150/80-100 (keep BP out of severe range $\geq 160/110$ but don't reduce MAP by >25% over 2h.

66. Navy Seal holding up a magazine

→ eclamptic seizure tx & prophylaxis = magnesium sulfate

67. "Heart stopping missions that will take your breath away!" headline on the magazine

→ magnesium toxicity = respiratory depression ($\downarrow$ DTRs prior), cardiac arrest.

Pts at $\uparrow$ risk → those

w/ renal insufficiency (Mg renally excreted). To reverse Mg toxicity → d/c MgSO₄, give Ca

68. Diver reading magazine drinking milk

→ magnesium toxicity = respiratory depression ($\downarrow$ DTRs prior), cardiac arrest.

Pts at $\uparrow$ risk → those

w/ renal insufficiency (Mg renally excreted). To reverse Mg toxicity → d/c MgSO₄, give Ca

69. Torpedo firing

→ preeclampsia spectrum definitive treatment = delivery!

70. "39" torpedo

→ CHTN window recommended time for delivery = 38-39wks GA

71. Smaller "37" torpedo with a clamp on it leaking high pressure steam

→ gHTN + preeclampsia w/o SF recommended time for delivery = 37wks GA

72. Smallest "34" torpedo with a clamp leaking even more high pressure steam

→ preeclampsia w/ SF + HELLP recommended time for delivery = 34wks GA

73. Moon face on smallest "34" torpedo

if delivering preeclampsia spectrum prior to 34 wks, remember to give antenatal corticosteroids

for fetal maturity

74. PPx badge + ASA vest + umpire mask

→ give daily low dose ASA to prevent preeclampsia spectrum in pts w/ → 1. H/o preeclampsia 2.

Multifetal gestation 3. cHTN 4. DM 5. Renal dz 6. SLE/APLS